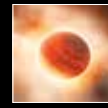




64-bit ARM for Post-K super

ISC Fujitsu has decided that it was use 64-bit ARMv8 cores for its 1000 PFLOPS super computer.
<http://dgit.in/postksuper>



Youngest exoplanet

The youngest known exoplanet discovered so far, K2-33b can teach us more about planet formation. <http://dgit.in/k233b>

Why Asteroseismology?

Stars have been guiding bodies for modern astrophysics. Much of what we know of our universe is from observing stars that pepper the multitude of galaxies. Most modern astronomical methods rely on de-constructing the light coming from stars. While classical methods have given us an idea of stellar physics, scientists have to rely on simplistic models to unravel the host of processes that dominate the world inside stars. Internally, stars are charac-

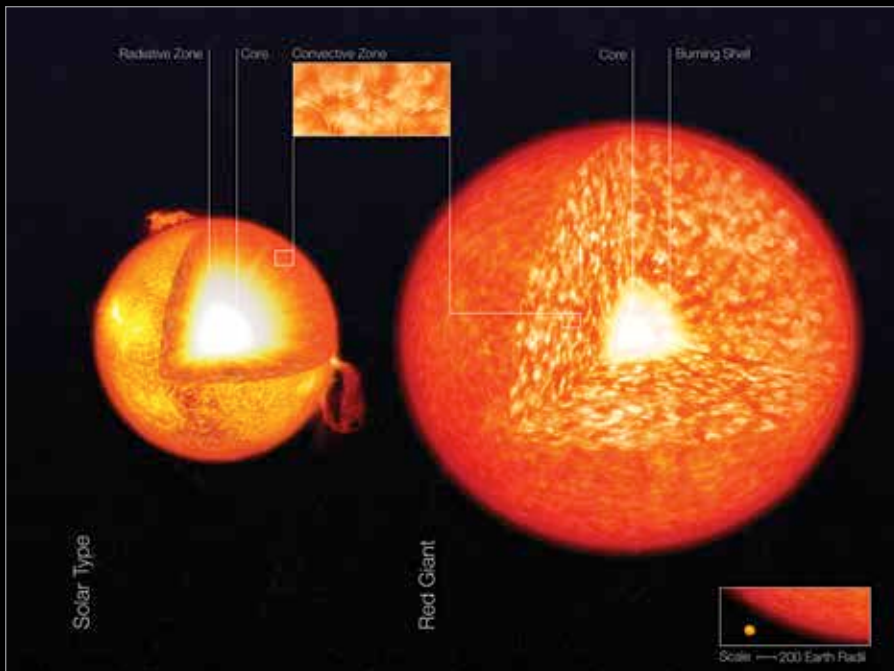
star - including its composition, structural properties and evolution.

What are stellar oscillations?

Stars, giant bodies of gas and plasma, usually exist in a state of equilibrium. The gravitational pull exerted on its elemental constituents is balanced by the massive amount of gas pressure inside the star. Any small perturbation however can cause a departure from equilibrium and lead to a state of oscillation - short lived

acoustic waves cannot travel through the vacuum in space (sound is longitudinal and needs a medium to propagate) they provide the perturbation that disturbs a star from its rest equilibrium state, causing it to oscillate.

Spherical bodies have three degrees of freedom and hence have multiple modes of oscillations, the simplest of them being radial modes. Radial modes manifest as a periodic expansion and contraction of a body in such a way that the stars spherical symmetry is conserved. The periodic displacement is hence equal in all directions. Imagine squishing a soft ball in such a way that the force used to compress and release is equally transmitted in all directions. At any time during this periodic motion, the ball retains its spherical shape with only its radius constantly increasing and decreasing with time. Non radial modes however don't preserve such a symmetry. The star oscillates in such a way that it loses its spherical shape. These non radial oscillations distort the star's surface in such a way that while some areas are expanding, some recede, hence leading to the formation of some exotic shapes. Stellar oscillations essentially refers to the distortion of a star's shape. Asteroseismology aims to study these oscillations in 'pulsating' stars and from it carry out a sort of reverse study to map the acoustic waves, find their origin and ultimately chart the composition of a star. Some GIF animations of the various shape modes can be found here : <http://dgit.in/Astroseismo>



Artist's impression showing the the radiation and convective zones in a star

terised by large scale convection, mixing, diffusion, and a host of chemical, hydrodynamic and magneto processes that are too complicated to be studied by approximate parameters used to model such processes. Simply put, most techniques offer poor precision. By studying stellar oscillations, the uncertainty in current models can be reduced significantly, thus providing scientists a viable diagnostic tool to study stars in their full glory. The inside of stars are inaccessible via traditional observation techniques. The advantage of using asteroseismology is that because stellar oscillations are almost uniquely a function of a star's properties, and different modes penetrate to different depths in a star, we can get high precision information about the various internal layers in a

or controlled. Imagine a block of wood connected to a spring system, left to fall under the influence of gravity. Instead of simply dropping the block, if we slowly guide the block and leave it, it reaches a certain height (its equilibrium height). Now if we disturb the block from this rest state, it begins to oscillate about that level. In real life, because the block will face air resistance, the motion gets damped and ultimately dies down. In an ideal scenario however the block continues to oscillate about its equilibrium position. Transfer this one dimensional picture to the three dimensional spherical shape of star, and it's not as easy to visualize these oscillations. Stars are far from being quiet bodies, with sound waves being generated in plenty due to its internal processes. While these

What can make a star oscillate?

There are different mechanisms that can drive the star to oscillate. One of the most popular explanations is the kappa mechanism that posits that certain layers within the sun have the ability to store energy. In the stars compression phase the layers store energy and release it when the star is in the expansion stage of its pulsation, leading to the star to overshoot its equilibrium radius. Energy is stored again when the surface recedes and the cycle continues. This driving mechanism is responsible for self excited oscillations in pulsating stars. Stars that are stable, are excited by virtue of their outer